

DTSA 5739

Security and Ethical Hacking: Attacking Web and AI Systems

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	Machine Learning Engineer (57.0 / 100)	46.8 / 100
SPECIALIZATION Cybersecurity for Data Science		

47

CURRENT AVG SCORE
across 6 job roles

78

POTENTIAL SCORE
with recommended additions

+38

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

In this last course of the sequence, students will learn how web application and server attacks are conducted against a variety of web technologies and frameworks. In addition, we will introduce the topic of Adversarial Machine Learning and attacks in the Artificial Intelligence domain, including Language Model systems. Students will gain an elementary understanding of the science and techniques behind these attacks, with an appropriate introduction to the AI world. Course assessments are through quizzes, hands-on exercises and an exam.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
Machine Learning Engineer	57.0	87.7	+30.8	HIGH
AI Engineer	48.1	82.6	+34.5	HIGH
ML Engineer	46.4	68.7	+22.3	MEDIUM
MLOps Engineer	45.2	78.2	+33.1	HIGH
Data Scientist	44.3	82.4	+38.1	HIGH
Applied Scientist	39.7	69.6	+29.9	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

✓ Course vocabulary is well-aligned with industry job descriptions.

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

✓ This course is already well-aligned with industry expectations. Continue monitoring for emerging tools and frameworks.

GAP KEYWORDS BY ROLE

Machine Learning Engineer	57.0 → 87.7 (+30.8 pts)
data tools exploratory design development models applications performance	
AI Engineer	48.1 → 82.6 (+34.5 pts)
perception data robotics models build technical problems technology	
ML Engineer	46.4 → 68.7 (+22.3 pts)
media data models build problems impact product new	
MLOps Engineer	45.2 → 78.2 (+33.1 pts)
detection data text media image tools models development	
Data Scientist	44.3 → 82.4 (+38.1 pts)
data product mitigation different social build people problems	
Applied Scientist	39.7 → 69.6 (+29.9 pts)
problems models technical data design development regulatory product	

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.